[S04] LIME and SHAP: Explaining Machine-Learning Predictions

Does a Highly Accurate Prediction Necessarily Come with a Reliable Explanation?

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Abstract

Post-hoc explanation methods are increasingly used to justify the predictions of accurate but opaque models in high-stakes domains such as medical diagnosis. Their use often rests on the implicit assumption that an accurate prediction comes with a reliable explanation. We examine this assumption for two widely used local explanation methods, LIME and SHAP. We first present both methods within a common framework of additive feature attribution and illustrate their objectives with hand-worked examples. We then compare LIME, KernelSHAP, and TreeSHAP on a random forest trained on the Breast Cancer Wisconsin dataset (97% test accuracy), using exact interventional TreeSHAP as the reference. The explanations are evaluated in terms of agreement, stability with respect to the random seed and the background data, runtime, and faithfulness measured by deletion curves. The methods agree on confidently classified instances but diverge in both feature ranking and sign on a borderline one. For a correctly classified borderline case, LIME produces six different top-ranked features across 30 random seeds, and changing the SHAP background reverses the sign of the two leading attributions. Moreover, a redundant noisy copy of `worst radius` receives as much attribution as the original. These results indicate that high accuracy does not guarantee a stable, unique, or causally meaningful explanation.

Keywords: explainable AI, LIME, SHAP, Shapley values, feature attribution, explanation stability, faithfulness.

1 Introduction

The most accurate machine learning models, such as deep neural networks and tree ensembles, are also among the hardest to interpret. In applications such as clinical oncology or credit risk assessment, accuracy alone is insufficient: a decision is acted upon only if its rationale can be examined and verified [1]. Explainable artificial intelligence (XAI) aims to provide such rationales, and among post-hoc local explanation methods, LIME [2] and SHAP [3] have become de facto standards in both research and practice.

An explanation, however, is itself the output of an algorithm with its own assumptions and hyperparameters. LIME fits a surrogate model to randomly sampled perturbations, and SHAP measures contributions relative to a chosen background distribution. In practice, it is often implicitly assumed that an accurate model yields explanations that are equally trustworthy. This paper examines that assumption.

1.1 Research Question

We address the question: *does a highly accurate prediction necessarily come with a reliable explanation?* We consider an explanation reliable if it is stable under the method's random choices and settings, faithful to the model it explains, and consistent across methods that claim to answer the same question.

Our contributions are as follows:

- We present LIME and SHAP within the common framework of additive feature attribution, define all quantities involved, and work through both methods by hand on the same illustrative model, which makes the absence of local accuracy in LIME directly visible.
- We compare LIME, KernelSHAP, and TreeSHAP on a random forest trained on the Breast Cancer Wisconsin dataset, using the exact attributions of interventional TreeSHAP as the reference.
- We quantify the sensitivity of each method to the random seed, the sample budget, and the background data, and identify concrete instances where explanations are unstable or contradictory.
- We evaluate faithfulness with deletion curves against a random baseline, and use correlated and duplicated features to illustrate why attributions must not be read as causal effects.

The experimental code and reproduction instructions are available on GitHub.

The remainder of the paper is organized as follows. Section 2 reviews the XAI taxonomy and the two methods. Section 3 formalizes LIME and SHAP, with a worked example in Section A. Section 4 reports our experiments, Section 5 discusses faithfulness and causality, and Section 6 concludes.

2 Background and Related Work

2.1 A Taxonomy of Explanation Methods

XAI methods are commonly organized along several dimensions [1, 4]. We summarize the four that are needed to position LIME and SHAP.

2.1.1 Interpretability and Explainability

Although the two terms are often used interchangeably, most of the literature distinguishes them [4]. *Interpretability* refers to the degree to which a human can directly follow how a model computes its output, as with a shallow decision tree or a sparse linear model. *Explainability* refers to the ability to provide, often through a separate method, a human-understandable account of why an opaque model produced a given output. In short, interpretability concerns the mechanism of a model, whereas explainability concerns the justification of its predictions. LIME and SHAP are explainability methods in this sense.

2.1.2 Intrinsic and Post-hoc Methods

Intrinsic (or ante-hoc) methods obtain transparency by construction, by restricting the model class to, e.g., shallow trees or generalized additive models [1]. *Post-hoc* methods leave an already trained model unchanged and construct an explanation of its behaviour afterwards. Both LIME and SHAP are post-hoc.

2.1.3 Local and Global Explanations

A *local* explanation accounts for a single prediction, for instance why a particular patient was assigned a high probability of malignancy. A *global* explanation describes the behaviour of the model as a whole, such as which features matter most across the population. LIME and SHAP are local methods. Local SHAP values can, however, be aggregated into global summaries, and LIME was complemented by a procedure for selecting representative local explanations [2].

2.1.4 Model-specific and Model-agnostic Methods

Model-specific methods exploit the internal structure of a model family and are typically efficient; TreeSHAP, for example, traverses the trees of an ensemble to compute Shapley values in polynomial time [5]. *Model-agnostic* methods require only the ability to query the model and therefore apply to any predictor. LIME and KernelSHAP are model-agnostic. In summary, LIME is a post-hoc, local, model-agnostic surrogate method; KernelSHAP is a post-hoc, local, model-agnostic attribution method; and TreeSHAP trades model agnosticism for exactness and speed on tree ensembles.

2.2 Local Surrogate Explanations: LIME

LIME was motivated by the problem of trust [2]. The authors distinguish whether a user should act on an individual prediction from whether a model as a whole should be deployed, and argue that held-out accuracy answers neither question reliably, since validation data may contain leakage or differ from deployment conditions. They require an explanation to be interpretable to its audience, locally faithful to the model, and model-agnostic, and they additionally call for a global perspective on the model. LIME addresses the first three requirements by fitting a sparse linear model to the behaviour of the black box on perturbed versions of the instance, weighted by their proximity to it (Section 3.1). For the global perspective, they propose SP-LIME, which greedily selects a small set of explanations that jointly cover the most important interpretable components, with a $(1-1/e)$ approximation guarantee owing to submodularity.

The evaluation combines simulated-user and human-subject experiments on text and image data. On sentiment classification of book and DVD reviews, explanations of sparse logistic regression and decision tree classifiers, which by construction use at most ten features, recovered more than 90% of these features, outperforming random, greedy, and Parzen-window baselines. On the 20 Newsgroups "Christianity vs. Atheism" task, where a classifier with 94% held-out accuracy relies on e-mail headers, non-expert users shown SP-LIME explanations identified the better-generalizing of two classifiers in 89% of cases, although held-out accuracy favoured the other, and used the explanations to remove spurious words and improve the classifier. In an image task deliberately biased towards snowy backgrounds, the number of participants who mentioned snow as a potential feature rose from 12 to 25 out of 27 after seeing explanations, while the number who trusted the flawed model fell from 10 to 3.

These results establish that LIME explanations can expose spurious correlations and support model selection. The evaluation, however, leaves several questions open. Faithfulness is measured only on models that are themselves sparse, for which a sparse linear explanation is well specified; neither the quality of the surrogate on non-linear models nor the variability of explanations across random seeds is quantified; and tabular data, for which the interpretable representation must be constructed by discretization, is not studied. The authors acknowledge that a linear surrogate cannot be faithful where the model is strongly non-linear even locally. Our experiments address these open questions on a tabular medical dataset.

2.3 Additive Feature Attribution: SHAP

By 2017, several local explanation methods, including LIME, DeepLIFT, and layer-wise relevance propagation, coexisted without a common theoretical basis for comparing them. Lundberg and Lee [3] showed that all of them belong to a single class of *additive feature attribution methods*, which explain a prediction as a sum of feature contributions. Drawing on cooperative game theory [6], they proved that within this class exactly one attribution satisfies three desirable properties, namely local accuracy, missingness, and consistency, and that this attribution is given by the Shapley values (Section 3.2). Since computing Shapley values exactly requires exponentially many model evaluations, they proposed KernelSHAP, a model-agnostic estimator obtained as a specially weighted linear regression, which can be seen as LIME with its heuristic

choices replaced by principled ones. Lundberg et al. [5] later proposed TreeSHAP, which computes exact Shapley values for tree ensembles in polynomial time.

3 Explanation Methods

3.1 Mathematical Formulation of LIME

3.1.1 Problem Setting

Let $f : \mathbb{R}^d \to \mathbb{R}$ denote the model to be explained and $x \in \mathbb{R}^d$ the instance of interest. In classification, $f(x)$ is the probability that x belongs to the class under consideration. An *explanation* of $f(x)$ is a model $g \in G$, where G is a class of potentially interpretable models, such as linear models or decision trees. LIME does not require g to approximate f over the entire input space; it requires only *local fidelity*, i.e., that g approximate f in a neighbourhood of x. This relaxation rests on the premise that a model whose global behaviour is too complex to be described by any $g \in G$ may nevertheless be well approximated by such a model locally [2]. The remainder of this section makes the notions of interpretability, neighbourhood, and approximation precise.

3.1.2 Interpretable Representation

Since an explanation must be intelligible to a human, g is defined not on the raw input features, e.g., word embeddings or pixel intensities, but on *interpretable components*, such as words (text), super-pixels, i.e., contiguous patches of similar pixels (images), or, for tabular data, statements of the form "feature j lies in the same value range as in x". LIME therefore represents x by a binary vector $x' \in \{0,1\}^{d'}$ that indicates the presence or absence of each interpretable component [2]; the number of components d' need not equal the input dimension d. Following the reference implementation, we define the components with respect to x itself, so that all of them are present in x and

$$x' = (1, 1, \ldots, 1). \tag{1}$$

The original paper describes the recovery of an input from its interpretable representation without naming the corresponding mapping. Following Lundberg and Lee [3], we denote it by $h_x : \{0,1\}^{d'} \to \mathbb{R}^d$. It converts any binary vector z' into a valid model input $z = h_x(z')$, for instance by deleting the absent words or greying out the absent super-pixels. The mapping is specific to x, since removing component j means removing it *from* x, and it satisfies $h_x(x') = x$.

3.1.3 Sampling for Local Exploration

Since g cannot be estimated from x alone, LIME draws a set $\mathcal{Z}$ of N *perturbed samples* around x'. Each sample z' is obtained by drawing non-zero elements of x' uniformly at random, where the number of draws is itself sampled uniformly, so that z' contains a fraction of the non-zero elements of x'. The sample is then mapped back to the input space:

$$x' \xrightarrow{\text{remove components}} z' \xrightarrow{h_x} z. \tag{2}$$

Since $x' = \mathbf{1}$, perturbation can only remove components. Each perturbed sample therefore has two representations: f is evaluated on z, as it is defined only on the original input space, whereas g is fitted on z', so that its coefficients refer to interpretable components. Table 1 illustrates this for a sentence with one component per word.

Table 1: A perturbed sample of a text instance in its two representations.

	Interpretable (g)	Model input (f)
Instance	$x' = (1, 1, 1, 1)$	"the food was great"
Perturbed sample	$z' = (1, 0, 1, 1)$	"the was great"

For tabular data, the reference implementation used in Section 4 discretizes every continuous feature into quartiles of the training distribution. A perturbed sample is generated by drawing, independently for each feature j, a quartile with probability equal to its frequency in the training data and then a value z_j inside that quartile. The component z'_j equals 1 if the drawn quartile coincides with that of x_j and 0 otherwise. Thus a removed component is not deleted but replaced by a value from another quartile, and even a retained component carries a new value from the quartile of x_j rather than x_j itself. Since each quartile has probability about 1/4, a typical perturbed sample retains only about a quarter of the components.

3.1.4 Optimization Objective

Local fidelity requires $g(z') \approx f(z)$, with greater emphasis on samples closer to x. Closeness is quantified by a *proximity measure*, defined as the exponential kernel

$$\pi_x(z) = \exp\left(-\frac{D(x,z)^2}{\sigma^2}\right), \tag{3}$$

where D is a distance function, e.g., cosine distance for text and L_2 distance for images, and $\sigma > 0$ is the kernel width. The weight lies in $(0, 1]$ and equals 1 at $z = x$. Adding a penalty $\Omega(g)$ on the complexity of g yields the LIME explanation

$$\xi(x) = \arg\min_{g \in G} \left[\mathcal{L}(f, g, \pi_x) + \Omega(g)\right], \tag{4}$$

with the locally weighted squared loss

$$\mathcal{L}(f, g, \pi_x) = \sum_{(z,z') \in \mathcal{Z}} \pi_x(z) \left(f(z) - g(z')\right)^2. \tag{5}$$

Ribeiro et al. restrict G to linear models $g(z') = w_g \cdot z'$, where $w_g \in \mathbb{R}^{d'}$ contains one coefficient per interpretable component, and use a hard sparsity constraint as the complexity penalty,

$$\Omega(g) = \infty \cdot \mathbb{I}\left[\|w_g\|_0 > K\right], \tag{6}$$

where $\|w_g\|_0$ is the number of non-zero coefficients and $\mathbb{I}[\cdot]$ the indicator function. The penalty is zero if g uses at most K components and infinite otherwise, so K is the length of the explanation, typically a small number such as 10. Note that Equation (4) does not enforce $g(x') = f(x)$; this observation is central to the comparison with SHAP in Section 3.2.

3.1.5 Fidelity–Interpretability Trade-off

Equation (4) balances two competing requirements: *local fidelity* to f, measured by $\mathcal{L}$, and *interpretability*, measured by Ω. The proximity measure π_x determines the

Require: classifier f, number of samples N
Require: instance x and its interpretable version x'
Require: similarity kernel π_x, length of explanation K
$\mathcal{Z} \leftarrow \emptyset$
for $i \in \{1, 2, \ldots, N\}$ **do**
  $z'_i \leftarrow \textsc{SampleAround}(x')$
  $z_i \leftarrow h_x(z'_i)$
  $\mathcal{Z} \leftarrow \mathcal{Z} \cup \{\langle z'_i, f(z_i), \pi_x(z_i)\rangle\}$
end for
$w \leftarrow \text{K-}\textsc{Lasso}(\mathcal{Z}, K)$
▷ z'_i as features, $f(z_i)$ as target
return w

Algorithm 1: Sparse linear explanations using LIME, adapted from [2].

locality of the explanation: by converting the distance from x into a weight, it ensures that nearby samples dominate the loss. The kernel width σ controls the size of the neighbourhood. A small σ yields a highly local fit that relies on few effectively weighted samples and therefore has high variance, whereas a large σ drives the surrogate towards a global linear approximation.

The penalty $\Omega(g)$ controls interpretability. Without it, g could assign non-zero weight to every component and would be no easier to interpret than f. Under Equation (6), the complexity penalty is equivalent to a constraint that selects at most K components.

3.1.6 Algorithm

Solving Equation (4) exactly under Equation (6) requires searching over all subsets of K components, which is intractable. Ribeiro et al. [2] approximate it with a procedure termed *K-LASSO*: K components are selected along the Lasso regularization path, and their weights are then learned by weighted least squares (Equation (5)). The complete procedure is given in Algorithm 1.

The cost is dominated by the N model evaluations and is independent of the size of the dataset. The tabular implementation used in our experiments deviates from the paper in minor respects. It fits a weighted ridge regression with an intercept, and it computes D as the Euclidean distance between z' and x', so that D^2 equals the number of removed components. Its default kernel width is $\sigma = 0.75\sqrt{d}$, where d is the number of input features, and it weights each sample by the square root of Equation (3), which corresponds to an effective width of $\sqrt{2}\,\sigma$.

The explanation consists of the K selected components and their signed coefficients. A positive coefficient indicates that the presence of the component supports the explained class, and a negative coefficient that it opposes it. For tabular data, a coefficient therefore describes the effect of the statement "x_j lies in its current quartile", not the effect of the raw value of x_j.

3.1.7 Limitations

Since $g(x')$ need not equal $f(x)$, LIME offers no local accuracy guarantee: the coefficients do not necessarily sum to the prediction being explained. The explanation also depends on the number of samples N, the kernel width σ, and the chosen interpretable representation (words, super-pixel segmentation, or value bins). Ribeiro et al. argue that weighting the samples by π_x makes LIME fairly robust to sampling noise [2]; our experiments in Section 4.3, however, show that explanations of borderline instances can change substantially across random seeds. Moreover, deleted words, greyed-out patches, or independently resampled feature values may lie far from the data distribution on which f was trained, so the surrogate may be fitted in regions where the behaviour of f is not representative of the data. Finally, as noted by Lundberg and Lee [3], the choices of π_x, $\mathcal{L}$, and Ω are heuristic. Section 3.2 shows that SHAP retains the same setting, a linear explanation over a binary interpretable representation, but replaces these choices with the unique ones implied by a small set of axioms.

3.2 Mathematical Formulation of SHAP

3.2.1 Additive Feature Attribution Methods

We retain the notation of Section 3.1: f is the model, x the instance, and g the explanation model. Following Lundberg and Lee [3], the interpretable representation is called the *simplified input* x', and $x = h_x(x')$. Local methods require $g(z') \approx f(h_x(z'))$ whenever $z' \approx x'$; the identity $h_x(x') = x$ holds even though x' contains less information than x, because h_x is specific to x.

Definition 1 (Additive Feature Attribution Methods)**.** *An explanation model g is an additive feature attribution method if it is defined as a linear function of binary variables:*

$$g(z') = \phi_0 + \sum_{i=1}^{M} \phi_i z'_i \tag{7}$$

where $z' \in \{0,1\}^M$ is a binary vector indicating the presence ($z'_i = 1$) or absence ($z'_i = 0$) of simplified feature i, M is the number of simplified input features, $\phi_i \in \mathbb{R}$ is the effect attributed to feature i, and $\phi_0 \in \mathbb{R}$ is a constant base value.

Such a method explains $f(x)$ as a base value plus the sum of the feature effects, evaluated at $x' = \mathbf{1}$. Lundberg and Lee [3] showed that widely adopted local explanation frameworks, including LIME, DeepLIFT, and Layer-Wise Relevance Propagation, all match this formulation, differing primarily in how they define the mapping h_x, the neighborhood kernel, and their specific attribution coefficients.

3.2.2 Shapley Values

In cooperative game theory, the Shapley value distributes the total payoff of a game among its players according to their average marginal contributions [6]. In the explanation setting, the features play the role of the players and the prediction that of the payoff, which yields the attribution

$$\phi_i(f, x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(M - |S| - 1)!}{M!} \times [f_x(S \cup \{i\}) - f_x(S)] \tag{8}$$

where $F = \{1, \ldots, M\}$ is the set of all features, S ranges over the subsets of F that exclude i, and the weight $|S|!\,(M - |S| - 1)!/M!$ equals the probability that, in a uniformly random ordering of the features, the features

preceding i are exactly those in S. The *value function* $f_x(S)$ is the expected prediction when only the features in S are known, and the base value is $\phi_0 = f_x(\emptyset) = \mathbb{E}[f(X)]$, where X denotes a random input drawn from the data distribution.

Since a model cannot in general be evaluated with features missing, Lundberg and Lee define the value of a coalition, $f_x(S) = f(h_x(z'))$ with $S = \{j : z'_j = 1\}$, as a conditional expectation and approximate it under successive simplifying assumptions:

$$f(h_x(z')) = \mathbb{E}[f(X) \mid X_S = x_S] \tag{9}$$
$$= \mathbb{E}_{X_{\bar{S}}|X_S=x_S}\left[f(x_S, X_{\bar{S}})\right] \tag{10}$$
$$\approx \mathbb{E}_{X_{\bar{S}}}\left[f(x_S, X_{\bar{S}})\right] \tag{11}$$
$$\approx f\left([x_S, \mathbb{E}[X_{\bar{S}}]]\right) \tag{12}$$

where x_S denotes the values of the features in S and $\bar{S} = F \setminus S$ the absent features. Equation (10) rewrites the conditional expectation as an expectation over the absent features, Equation (11) assumes feature independence, and Equation (12) additionally assumes model linearity. KernelSHAP and interventional TreeSHAP estimate Equation (11) with a background set, whereas Equation (12) corresponds to replacing absent features by their means.

3.2.3 Properties and Uniqueness

Lundberg and Lee [3] require an additive attribution to satisfy three properties. The first, *local accuracy* (efficiency), requires the explanation model to match the original model at the explained instance:

$$f(x) = g(x') = \phi_0 + \sum_{i=1}^{M} \phi_i x'_i, \tag{13}$$

so that, since $x' = \mathbf{1}$, the attributions plus the base value $\phi_0 = f_x(\emptyset)$ recover the prediction exactly. The second, *missingness*, requires features absent from the simplified input to receive no attribution:

$$x'_i = 0 \implies \phi_i = 0. \tag{14}$$

The third, *consistency* (monotonicity), states that if the marginal contribution of feature i in a model f' is never smaller than in a model f,

$$f'_x(S \cup \{i\}) - f'_x(S) \geq f_x(S \cup \{i\}) - f_x(S) \quad \forall S \subseteq F \setminus \{i\}, \tag{15}$$

then its attribution cannot decrease, i.e., $\phi_i(f', x) \geq \phi_i(f, x)$.

Theorem 1 (Uniqueness of Shapley Attributions)**.** *Only one explanation model g follows Definition 1 and satisfies local accuracy, missingness, and consistency: the attributions $\phi_i(f, x)$ are given by the Shapley values in Equation (8).*

Consequently, for a given mapping h_x, any additive attribution method not based on Shapley values violates local accuracy or consistency, since missingness is satisfied by most methods.

3.2.4 Model-Agnostic Approximation: KernelSHAP

Direct computation of Equation (8) requires evaluating 2^M coalitions; for the 30 features of Breast Cancer Wisconsin this amounts to $2^{30} \approx 10^9$ evaluations per explanation, which is impractical.

The weighted regression of LIME (Equation (4)) and the combinatorial Shapley formula (Equation (8)) appear unrelated. However, linear LIME is an additive feature attribution method, so by Theorem 1 its heuristic choices of $\mathcal{L}$, π_x, and Ω in general violate local accuracy or consistency. Lundberg and Lee [3] solved the inverse problem, deriving the choices under which the regression recovers the Shapley values:

Theorem 2 (Shapley Kernel)**.** *Under Definition 1, the unique choices of regularization penalty $\Omega(g)$, weighting kernel $\pi_x(z')$, and squared loss function $\mathcal{L}$ that ensure that the solution of the surrogate objective equals the Shapley values are:*

$$\Omega(g) = 0 \tag{16}$$
$$\pi_x(z') = \frac{M-1}{\binom{M}{|z'|}\,|z'|\,(M-|z'|)} \tag{17}$$
$$\mathcal{L}(f, g, \pi_x) = \sum_{z' \in \mathcal{Z}} \left[f(h_x(z')) - g(z')\right]^2 \pi_x(z') \tag{18}$$

where $|z'| = \sum_{j=1}^{M} z'_j$ denotes the number of non-zero (active) features in binary coalition vector z'.

The SHAP kernel in Equation (17) satisfies $\pi_x(z') \to \infty$ when $|z'| \in \{0, M\}$ (i.e., for the null coalition $\mathbf{0}$ and the grand coalition $\mathbf{1}$). These infinite weights act as strict equality constraints on the optimization problem:

$$g(\mathbf{0}) = \phi_0 = f_x(\emptyset), \qquad g(\mathbf{1}) = \phi_0 + \sum_{i=1}^{M} \phi_i = f(x) \tag{19}$$

which enforce local accuracy exactly. In practice, the infinite weights are replaced by fixing ϕ_0 and imposing the second equality as a linear constraint.

Theorem 2 expresses the Shapley values as the solution of a weighted least-squares problem over all 2^M coalitions. KernelSHAP approximates this problem by sampling. It draws N coalitions $z'_1, \ldots, z'_N \in \{0,1\}^M \setminus \{\mathbf{0}, \mathbf{1}\}$ with probability proportional to $\pi_x(z')$ and estimates the value of each coalition over a background set B of reference instances,

$$\hat{f}_x(S_k) = \frac{1}{|B|} \sum_{r \in B} f\left(x_{S_k}, r_{\bar{S}_k}\right), \quad S_k = \{j : z'_{kj} = 1\}, \tag{20}$$

which is the empirical counterpart of Equation (11). With $\phi_0 = \hat{f}_x(\emptyset)$, the attributions are then obtained as

$$\hat{\phi} = \arg\min_{\phi} \sum_{k=1}^{N} \left(\hat{f}_x(S_k) - \phi_0 - \sum_{i=1}^{M} \phi_i z'_{ki}\right)^2 \quad \text{s.t.} \quad \sum_{i=1}^{M} \phi_i = f(x) - \phi_0. \tag{21}$$

Since the coalitions are already drawn according to the kernel, they enter the regression with equal weights, and

the constraint is handled by eliminating one coefficient. Each explanation requires $N \cdot |B|$ model evaluations. Lundberg and Lee [3] report that this joint regression attains a given accuracy with considerably fewer evaluations than estimating each attribution separately by permutation sampling.

3.2.5 Limitations

Interventional SHAP attributions depend on the background distribution used to represent absent features. When features are correlated, combining retained values from the explained instance with values from background rows can produce unlikely feature combinations. In Section 4.3, changing the background reverses the signs of the two leading attributions for the borderline instance.

KernelSHAP introduces sampling variability when it estimates attributions from a finite set of coalitions. The local accuracy constraint ensures that the attributions sum to the difference between the prediction and the baseline, even when individual estimates vary across runs. In Section 4.3, with 2,000 coalitions, the same five features appeared in the top five of every run for the borderline instance, but two different features ranked first across runs.

3.3 Illustrative Example

Section A computes both explanations by hand for the two-feature model $f(x_1, x_2) = 0.2 + 0.4x_1 + 0.3x_2 + 0.2x_1x_2$ at $x = (1, 1)$. The Shapley values $\phi_1 = 0.275$ and $\phi_2 = 0.225$ add up exactly to $f(x) - \phi_0 = 0.50$, whereas the LIME surrogate ranks the features identically but yields $g(x') = 1.096 \neq f(x) = 1.10$, since a linear model cannot represent the interaction term. The example thus shows on a minimal case that the difference between the two methods lies in the choice of kernel, which in SHAP enforces local accuracy.

4 Experimental Evaluation

We evaluate the explainers in two experiments: the first compares the explanations produced for individual predictions, and the second measures their sensitivity to random seeds, sample budgets, and background data. Faithfulness and causality are examined in Section 5. Implementation details are given in Section B.

4.1 Experimental Setup

4.1.1 Dataset

The experiments use the Breast Cancer Wisconsin (Diagnostic) dataset [7], which contains 569 instances described by 30 continuous features computed from digitized images of fine-needle aspirate biopsies. The features describe ten characteristics of the cell nuclei (radius, texture, perimeter, area, etc.), each summarized by its mean, standard error, and "worst" (largest) value. We encode the target as malignant ($y = 1$, 212 instances) versus benign ($y = 0$, 357 instances), so that a positive attribution always pushes the prediction towards malignancy. The data are split into stratified training (455) and test (114) sets with a fixed seed. Several features are almost redundant; for example, the mean radius, perimeter, and area have pairwise correlations of about 0.99. This redundancy plays a central role in Section 5.

4.1.2 Models

The explained model is a random forest (RF) with 300 trees, which reaches a test accuracy of 0.974 (ROC AUC 0.994, 3 errors out of 114); the explained output is $f(x) = P(\text{malignant} \mid x)$. Since f is a tree ensemble, interventional TreeSHAP computes its SHAP values exactly with respect to the background set, and we use it as the deterministic reference for both LIME and KernelSHAP.

4.1.3 Explainers and Settings

Every feature is an interpretable component, so $d' = M = d = 30$. LIME runs in regression mode on f, with quartile discretization, $N = 5000$ samples, $K = 30$ (no sparsity constraint, so that all methods rank the same 30 features), and the library's default kernel width $\sigma = 0.75\sqrt{d} \approx 4.1$. KernelSHAP uses 2000 coalition samples. Both SHAP variants share a background set B of 100 random training instances. As a sanity check, on an RF refitted on three features, KernelSHAP and interventional TreeSHAP reproduce to three decimals the Shapley values obtained by enumerating all eight coalitions, as in Section A.

4.1.4 Selected Instances

Three test instances are chosen from the RF probabilities to cover different levels of model confidence: the most confident correct benign and malignant predictions ($P(\text{malignant}) = 0.000$ and 1.000), and a *borderline* malignant tumour that RF classifies correctly with $P(\text{malignant}) = 0.513$.

4.2 Experiment 1: Local Explanations and Agreement

4.2.1 Feature Contributions

Figure 1 shows the ten largest attributions of each method for the three instances, and Figure 2 decomposes the borderline prediction additively from the base value to $f(x)$.

4.2.2 LIME-versus-SHAP Comparison

Table 2 compares KernelSHAP and LIME with the reference. *Jaccard@5* is the overlap of the top-5 feature sets; ρ is the Spearman rank correlation of $|\phi_j|$ over all 30 features; *Sign@5* is the share of the reference's top-5 features with matching sign; *Gap* is $|\tilde{f}(x) - f(x)|$, where the reconstruction $\tilde{f}(x)$ is $\phi_0 + \sum_j \phi_j$ for SHAP and $g(x')$ for LIME; and R^2 is the coefficient of determination of the surrogate g on its perturbed samples, weighted by π_x. KernelSHAP and interventional TreeSHAP estimate the same quantity, Equation (11) over the same background set, so they must agree as $N \to \infty$; with $N = 2000$ of the 2^{30} possible coalitions, however, the agreement is an empirical question. The KernelSHAP gap, in contrast, is zero by construction, because the regression enforces $\sum_j \phi_j = f(x) - \phi_0$.

4.2.3 Discussion

With 2000 coalitions, KernelSHAP recovers the reference top-5 set with matching signs on every instance; its lower rank correlation on the confident malignant instance (0.89) stems from the ordering of the less important features, as no attribution deviates from the reference by more than 0.004. Both SHAP estimators reconstruct $f(x)$ exactly, whereas the LIME surrogate deviates from $f(x)$ by up to

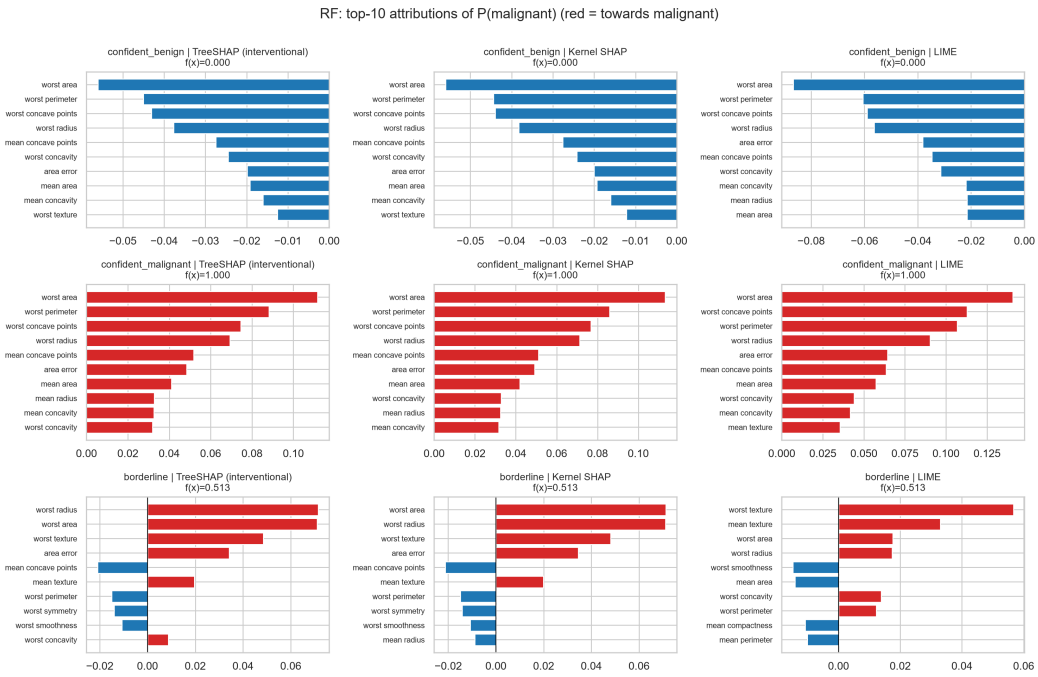


Figure 1: Top-10 attributions of $P(\text{malignant})$ for the three instances (rows) by TreeSHAP (interventional), KernelSHAP, and LIME (columns). Red bars push towards malignant, blue bars towards benign.

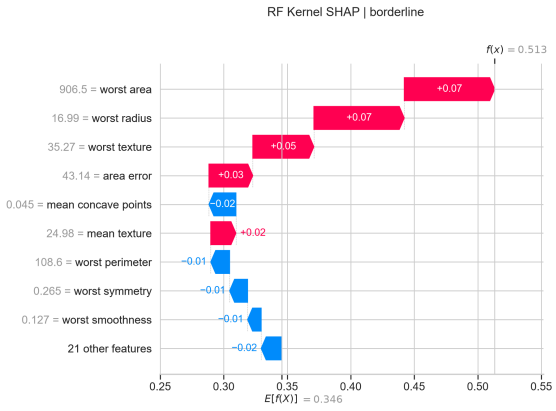


Figure 2: KernelSHAP waterfall for the borderline RF instance: the attributions add up exactly from $\mathbb{E}[f(X)] = 0.346$ to $f(x) = 0.513$.

Table 2: Agreement of KernelSHAP (KS) and LIME with the reference (interventional TreeSHAP).

Instance	Method	Jacc.@5	ρ	Sign@5	Gap	R^2
Conf. benign	KS	1.00	1.00	1.0	0.000	–
	LIME	0.67	0.80	1.0	0.100	0.22
Conf. malig.	KS	1.00	0.89	1.0	0.000	–
	LIME	0.67	0.85	1.0	0.056	0.68
Borderline	KS	1.00	0.99	1.0	0.000	–
	LIME	**0.43**	**0.67**	**0.6**	0.019	0.06

0.10 in probability, with a weighted R^2 between 0.06 and 0.68; the local linear model thus often fails to approximate f even near x.

Agreement between the methods depends on the confidence of the prediction. For the two confident instances, LIME and SHAP share four of their top-5 features with identical signs. For the borderline instance, LIME shares only three top-5 features with the reference and reverses the sign of two of them, `area error` and `mean concave points`. LIME ranks `worst texture` and `mean texture` first, whereas SHAP attributes the prediction mainly to `worst radius` and `worst area`, to which LIME assigns only a quarter of their SHAP value (Figure 1, bottom row). Part of this disagreement arises because the two methods measure the effect of x_j against different baselines. A LIME coefficient measures how much the surrogate changes when feature j lies in the same quartile as x_j rather than in another quartile, and it is not constrained to sum to any particular value. A SHAP value measures how much the expected prediction changes when the exact value x_j is known rather than averaged over the background set, and the SHAP values sum to $f(x) - \phi_0$. Both indicate whether x_j pushes the prediction above or below a typical patient, so their signs and ranks can be compared, but their magnitudes cannot: on the borderline instance, for example, `worst radius` receives 0.017 from LIME and 0.071 from SHAP.

4.3 Experiment 2: Explanation Stability

4.3.1 Sensitivity to the Random Seed and Sample Budget

LIME is run with 30 seeds for each $N \in \{500, 1000, 5000\}$, and KernelSHAP with 15 seeds for each $N \in \{100, 500, 2000\}$, on all three instances. Stability is measured by the mean pairwise Jaccard@5 between runs, the number of distinct top-1 features, and the number of

features whose sign changes across runs (Table 3).

Table 3: Seed stability on RF. "Top-1" counts distinct top-1 features across runs; "Flips" counts features whose sign changes.

Instance	Method	N	Jacc.@5	Top-1	Flips
Conf. malig.	LIME	500	0.79	1	0
Borderline	LIME	500	**0.25**	**6**	**28**
Borderline	LIME	5000	0.54	1	8
Borderline	KernelSHAP	100	0.92	2	0
Borderline	KernelSHAP	2000	1.00	2	0

The borderline RF instance provides a clear example of an unstable explanation. Although the model classifies it correctly, LIME with $N = 500$ produces six different top-1 features across 30 seeds, and 28 of the 30 features change sign at least once. The mean pairwise Jaccard@5 of 0.25 corresponds to only two shared features out of five. On the same instance, KernelSHAP keeps the same top-5 set in every run once $N \geq 500$, and its mean absolute deviation from TreeSHAP decreases from 1.3×10^{-3} to 1.9×10^{-4} as N grows from 100 to 2000. Varying LIME's kernel width over $\{0.25, 0.75, 3\} \cdot \sqrt{d}$ at $N = 5000$ changes the borderline Jaccard@5 only from 0.53 to 0.55, so on this dataset the sample budget matters far more than σ.

4.3.2 Sensitivity to the Background Set

SHAP values are defined relative to a background distribution. Recomputing the explanations with five different backgrounds changes the base value ϕ_0 substantially (Table 4). Under the malignant-only background, `worst area` and `worst radius`, the two leading features of the borderline instance, *reverse sign*. Both readings are correct: compared with an average patient, the tumour is large; compared with a typical malignant tumour, it is small. Figure 3 in Section C shows the attributions under all five backgrounds: the two random backgrounds and the k-means summary give nearly identical attributions, whereas under the malignant-only background the ranking also changes, with `worst perimeter` receiving the largest, negative, attribution.

Table 4: Base value ϕ_0 under different backgrounds, and the number of the borderline RF instance's top-5 features whose sign flips relative to the default.

Background	ϕ_0	Flips
Random 100 (default)	0.346	0
Random 20	0.296	0
k-means, 10 centroids	0.334	0
Benign only	0.027	1
Malignant only	0.956	**3**

4.3.3 Discussion

All methods are stable on confident instances, but the explanation of the borderline instance, precisely the kind of case that most needs scrutiny, is markedly less stable. LIME is more sensitive to randomness than KernelSHAP: KernelSHAP converges to a well-defined target as N grows, whereas the LIME solution depends on π_x and on the discretization, and even at $N = 5000$ the borderline Jaccard@5 reaches only 0.54. This stability comes at a cost. At $N = 2000$, KernelSHAP is four to five times slower than LIME at $N = 5000$ (≈ 0.7–0.9 s versus ≈ 0.17 s), because every coalition is averaged over the 100 background rows, while TreeSHAP takes ≈ 0.02 s. SHAP, in turn, is sensitive to the reference: the background moves ϕ_0 from 0.03 to 0.96 and changes attribution signs, so it is an integral part of a SHAP explanation and must be reported with it.

4.4 Comparison Summary

Table 5 summarizes the three explainers, combining the theory of Section 3 with the measurements above.

5 Discussion

5.1 Evaluating Faithfulness: Deletion Curves

An explanation is *faithful* if the features it ranks highest are those the model actually relies on. We propose and apply the following procedure:

1. For each instance and explainer, rank the features by their attribution towards the predicted class.
2. For $k = 0, 1, \ldots, 30$, replace the top-k features with values drawn from the background set (50 rows) and record the mean probability of the predicted class.
3. Summarize each curve by its normalized area under the curve (AUC). A faithful ranking makes the probability fall quickly and yields a low AUC.
4. Compare against a random feature ordering (mean of 10 permutations) as the baseline.

The procedure is applied to 40 test instances (20 per class).

All three explainers are far more faithful than the random baseline, and SHAP is slightly ahead of LIME (Table 6). This advantage should not be overstated: the test replaces features with background values, which is exactly the notion of feature absence underlying interventional SHAP, and the difference of 0.03 in AUC is measured on only 40 instances.

5.2 The Causality Caveat: Feature Importance ≠ Causal Effect

Attribution scores describe *what the model relies on*, not *what causes the diagnosis*. To illustrate this, we retrain RF on two modified feature sets. In the *reduced* set, the mean and worst perimeter and area are removed, as they are nearly deterministic functions of the radius. In the *copy* set, a duplicate of `worst radius` with Gaussian noise (5% of its standard deviation; correlation 0.999) is added.

As Table 7 shows, after retraining without the perimeter and area features, the mean absolute TreeSHAP attribution of `worst radius` increased from 0.041 to 0.090, while test errors increased from three to five out of 114 cases. After retraining with a redundant noisy copy of `worst radius`, the original and the copy received similar mean absolute TreeSHAP attributions (0.039 and 0.040, respectively). The original and augmented models each made three test errors. Equal error counts alone do not establish that the models made the same predictions on individual cases. Both LIME and interventional SHAP also vary features independently and thus evaluate the model on implausible inputs, such as a large radius combined with a small area. How credit is split among correlated features

Table 5: Summary comparison of LIME, KernelSHAP, and TreeSHAP (theory and our measurements).

Dimension	LIME	KernelSHAP	TreeSHAP
Foundation	Locally weighted sparse linear surrogate	Shapley values via weighted linear regression	Shapley values via tree traversal
Uniqueness	Heuristic choices of π_x, Ω, $\mathcal{L}$	Unique solution under three properties	Unique solution under three properties
Local accuracy	Not guaranteed (gap up to 0.10)	Exact (gap ≈ 0)	Exact (gap $\approx 10^{-4}$)
Model scope	Any model	Any model	Tree ensembles only
Determinism	Stochastic	Stochastic, converging in N	Deterministic
Agreement on the borderline instance	Low (Jaccard@5 0.43)	High (Jaccard@5 1.00)	Reference
Main sensitivity	Seed, N, discretization	Background set, N	Background set
Runtime per explanation	≈ 0.15 s	≈ 0.7–0.9 s	≈ 0.02 s
Deletion AUC (lower is better)	0.600	0.574	0.574

Table 6: Deletion AUC over 40 test instances (lower is more faithful).

Method	AUC
KernelSHAP	**0.574**
TreeSHAP	**0.574**
LIME	0.600
Random	0.743

Table 7: Attribution of `worst radius` and its noisy copy under the three feature sets (global: mean $|\phi_j|$ of TreeSHAP over the test set; local: confident malignant instance). Test accuracy of RF in the last column.

Feature	Variant	Global TreeSHAP	Local TreeSHAP	Local LIME	Acc.
worst radius	full	0.041	0.069	0.090	0.974
worst radius	reduced	**0.090**	**0.157**	**0.213**	0.956
worst radius	+ copy	0.039	0.069	0.086	0.974
noisy copy	+ copy	**0.040**	**0.070**	**0.095**	0.974

therefore depends on the model and the explainer, not on the underlying biology. Establishing a causal effect requires an interventional study or an explicit causal model; it cannot be inferred from an attribution ϕ_j.

5.3 Limitations of This Study

The experiments use a single dataset, a single train/test split, a single explained model, three instances for the local analysis, and 40 instances for the deletion test. They are therefore illustrative rather than conclusive. Moreover, an explanation describes one particular model: equally accurate models may explain the same patient differently (the Rashomon effect), which we do not examine. Only LIME's default tabular configuration was examined, and the deletion test shares its notion of feature absence with interventional SHAP.

6 Conclusion

This paper asked whether a highly accurate prediction necessarily comes with a reliable explanation. Within the scope of our experiments, the answer is negative. High accuracy reflects average performance over a test distribution and guarantees neither the stability, the uniqueness, nor the causal validity of a post-hoc explanation. Our random forest reaches about 97% accuracy, yet for a correctly classified borderline instance LIME with 500 samples produced six different top-ranked features across 30 random seeds. LIME and SHAP agreed on confident instances but disagreed in ranking and sign on the borderline one, which is precisely the kind of case in which an explanation is most needed. Changing the SHAP background moved the base value of the random forest from 0.03 to 0.96 and reversed the sign of the leading attributions. Removing the perimeter and area features more than doubled the mean absolute TreeSHAP attribution of `worst radius`, while test accuracy decreased from 97.37% to 95.61%. In a separate experiment, adding a redundant noisy copy of `worst radius` left test accuracy at 97.37%, and the copy received nearly the same mean absolute TreeSHAP attribution as the original feature.

These findings suggest several practical guidelines. For tree ensembles, TreeSHAP is preferable, as it is exact, deterministic, and orders of magnitude faster than the model-agnostic estimators; the background set and the feature-perturbation mode should be reported with the explanation. In model-agnostic settings, KernelSHAP should be preferred when local accuracy matters. When LIME is used, explanations should be repeated over several seeds with a budget of at least several thousand samples, and the R^2 of the surrogate should be reported. In all cases, explanations should be validated with a deletion test against a random baseline, and statistical attribution should be clearly distinguished from causal effect when results are communicated to clinicians or other stakeholders.

Our study is limited to a single dataset and a single explained model. Future work could extend the analysis to further datasets and model families, compare against explainers that respect feature dependence, such as conditional or causal Shapley values, and develop standardized faithfulness benchmarks for medical applications.

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A Worked Example

We compute the SHAP and LIME explanations by hand for the two-feature model

$$f(x_1, x_2) = 0.2 + 0.4x_1 + 0.3x_2 + 0.2x_1x_2, \qquad (22)$$

where X_1 and X_2 are independent and uniformly distributed on $[0, 1]$, so that $\mathbb{E}[X_1] = \mathbb{E}[X_2] = 0.5$. The instance is $x = (1, 1)$, with prediction $f(1, 1) = 1.10$.

A.1 SHAP Values

The base value is

$$\begin{aligned}\phi_0 = \mathbb{E}[f(X)] &= 0.2 + 0.4(0.5) + 0.3(0.5) + 0.2(0.5)(0.5) \\ &= 0.60.\end{aligned} \qquad (23)$$

For feature 1, the coalitions are $S = \emptyset$ and $S = \{2\}$, both with weight $\frac{0!\,1!}{2!} = \frac{1!\,0!}{2!} = \frac{1}{2}$. The required values are

$$\begin{aligned}f_x(\{1\}) &= \mathbb{E}[f(1, X_2)] = 0.2 + 0.4 + 0.15 + 0.1 = 0.85, \\ f_x(\{2\}) &= \mathbb{E}[f(X_1, 1)] = 0.2 + 0.2 + 0.3 + 0.1 = 0.80,\end{aligned}$$

so the marginal contributions of feature 1 are $0.85 - 0.60 = 0.25$ for $S = \emptyset$ and $1.10 - 0.80 = 0.30$ for $S = \{2\}$, and

$$\phi_1 = \tfrac{1}{2}(0.25) + \tfrac{1}{2}(0.30) = 0.275. \qquad (24)$$

Similarly,

$$\phi_2 = \tfrac{1}{2}(0.80 - 0.60) + \tfrac{1}{2}(1.10 - 0.85) = 0.225, \qquad (25)$$

and local accuracy holds: $\phi_0 + \phi_1 + \phi_2 = 0.60 + 0.275 + 0.225 = 1.10 = f(1, 1)$.

A.2 LIME Explanation

Each feature is one interpretable component, so $x' = (1, 1)$. A removed component is replaced by its mean 0.5, i.e., $[h_x(z')]_j = x_j$ if $z'_j = 1$ and 0.5 otherwise. Because f is linear in each feature separately and the features are independent, this mean imputation (Equation (12)) yields the same coalition values as the expectations used above (Equation (11)), so the two explanations are based on identical information. With $d' = 2$, we enumerate all four perturbed samples instead of sampling them, take $D(x, z)^2$ as the number of removed components, and set $\sigma = 1$, so that $\pi_x = e^{-D^2}$:

z'	$z = h_x(z')$	$f(z)$	$\pi_x(z)$
$(1, 1)$	$(1, 1)$	1.10	1.000
$(1, 0)$	$(1, 0.5)$	0.85	0.368
$(0, 1)$	$(0.5, 1)$	0.80	0.368
$(0, 0)$	$(0.5, 0.5)$	0.60	0.135

With $K = 2$ the sparsity constraint is inactive. Following the reference implementation, the surrogate includes an intercept, $g(z') = w_0 + w_1 z'_1 + w_2 z'_2$, and is obtained by weighted least squares, $w = (Z^\top W Z)^{-1} Z^\top W y$, where Z has rows $(1, z'_1, z'_2)$, $W = \mathrm{diag}(\pi_x)$, and $y = f(z)$. This gives

$$w_0 = 0.573, \qquad w_1 = 0.287, \qquad w_2 = 0.237, \qquad (26)$$

and hence $g(x') = 1.096$.

A.3 Comparison

The ranking agrees with SHAP, but local accuracy fails: $g(x') = 1.096 \neq f(x) = 1.10$, because the interaction term $0.2\,x_1x_2$ cannot be represented by a linear g and the residual is spread over all four samples. The result also depends on σ. As $\sigma \to \infty$, all weights become equal and the fit reduces to ordinary least squares, giving $w_1 = 0.275$, $w_2 = 0.225$, $w_0 = 0.5875 \neq \phi_0$, and $g(x') = 1.0875$, so local accuracy still fails. The slopes coincide with the Shapley values only because $M = 2$: an unweighted regression over all coalitions yields the Banzhaf values, which equal the Shapley values for two features but differ from them in general. The SHAP kernel (Equation (17)) instead assigns infinite weight to $z' = \mathbf{0}$ and $z' = \mathbf{1}$, which forces $g(\mathbf{0}) = \phi_0$ and $g(\mathbf{1}) = f(x)$ and recovers the Shapley values for any M.

B Reproducibility and Source Code

B.1 Computational Environment

All experiments were executed in the following environment:

- **Language:** Python 3.12.
- **Core libraries:** `scikit-learn` 1.9, `numpy` 2.5, `pandas` 3.1, `scipy`.
- **Explainability libraries:** `shap` 0.54, `lime` 0.2.
- **Visualization:** `matplotlib`, `seaborn`.

B.2 Source Code

The experiments are implemented in three independent, deterministic scripts sharing a helper module:

- `exp1_local_explanations.py`: three-feature Shapley sanity check, local explanations, and the comparison table (Section 4.2).
- `exp2_stability.py`: sensitivity to the seed, sample budget, kernel width, and background (Section 4.3).
- `exp3_faithfulness.py`: deletion curves and the correlated-feature experiment (Section 5).
- `common.py`: data split, models, instance selection, explainer wrappers, and metrics.

B.3 Hyperparameters and Configurations

- **Data split:** stratified 80/20, `random_state=42`; label 1 = malignant.
- **Random Forest:** `n_estimators=300`, `random_state=42`, other parameters at their defaults.
- **Background set:** 100 random training instances, shared by KernelSHAP and TreeSHAP.
- **LIME:** regression mode, quartile discretizer, `num_samples=5000`, `kernel_width` $= 0.75\sqrt{d}$ with $d = 30$, all 30 features retained.
- **KernelSHAP:** `nsamples=2000`, `l1_reg=False`.
- **TreeSHAP:** interventional feature perturbation with probability output (reference); tree-path-dependent perturbation for comparison.

C Attributions under Different Backgrounds

Figure 3 complements Table 4 by showing the KernelSHAP attributions of the borderline instance under each of the five backgrounds, for the eight features with the largest attributions under the default background.

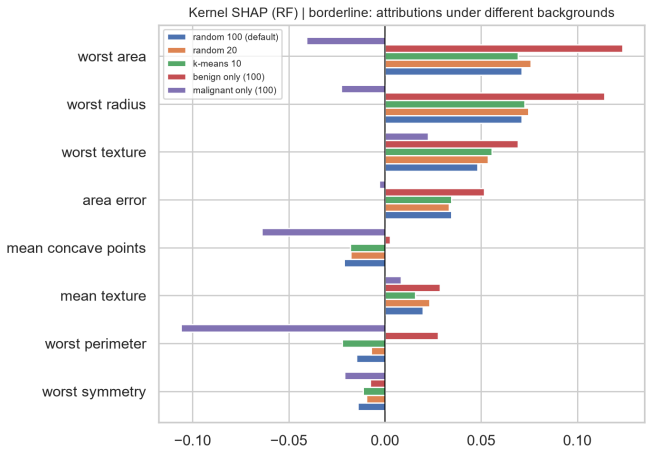


Figure 3: KernelSHAP attributions of the borderline RF instance under five background sets.